R15

Code No: 121AL

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech I Year Examinations, October/November - 2020
MATHEMATICAL METHODS
(Common to EEE, ECE, CSE, EIE, IT, ETM)

Time: 2 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1. Using Newton's forward formula compute $f(142)$ from the following table:

x	140	150	160	170	180
$f(x)$	3.685	4.854	6.302	8.076	10.225

[15]

2. By the method of least squares, find the straight line that best fits in the following data:

x:	1	2	3	4	5
y:	14	27	40	55	68

[15]

3.a) Determine a root of $xe^x - 2 = 0$ by method of false position.

b) Determine $\frac{dy}{dx}$ *and* $\frac{d^2y}{dx^2}$ at $x = 0$ from the following data

x	0	1	2	3	4	5
y	4	8	15	7	6	2

[7+8]

4.a) Given that $\frac{dy}{dx} = 2 + \sqrt{xy}$, $y(1) = 1$. Find $y(2)$ in steps of 0.2 using the Euler's method.

b) Evaluate $\int_0^{0.6} e^{-x^2} dx$ by using Simpson's $\frac{1}{3}$ rd rule taking seven ordinates. [8+7]

5. Find the half – range sine and cosine series expansions of the function

$$f(x) = \begin{cases} \frac{2k}{L}x, & if \quad 0 < x < \frac{L}{2} \\ \frac{2k}{L}(L-x), & if \quad \frac{L}{2} < x < L \end{cases}.$$

[15]

6. Find the Fourier transform of $f(x) = \begin{cases} 1-x^2 & for \quad |x| \le 1 \\ 0 & for \quad |x| > 1 \end{cases}$ hence evaluate $\int_0^{\infty} \left(\frac{x\cos x - \sin x}{x^3} \right) \cos\frac{x}{2}\, dx$.

[15]

7.a) Derive the partial differential equation by eliminating the constants a and b from the equation $z = ax^2 + by^2$.

b) Solve the partial differential equation $(x^2 - yz)p + (y^2 - zx)q = z^2 - xy$. [7+8]

8. State Gauss divergence and Verify Gauss divergence theorem for the function $\bar{F} = 4x\,\bar{i} - 2y^2\,\bar{j} + z^2\,\bar{k}$ over the cylindrical region bounded by $x^2 + y^2 = 4$, $z = 0$ *and* $z = 3$.

[15]

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TUH USED PAPERS OCT/NOV